MEESHO WOMEN'S FASHION E-COMMERCE DATABASE MANAGEMENT SYSTEM

SQL QUERY IMPLEMENTATION

1. Basic SQL Queries

1.1 SELECT Query

The SELECT statement is used to retrieve data from a table.

Display all products:

```
SQL> SELECT * FROM Meesho_Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID     PRICE STOCK_QUANTITY
----------- ---------- --------------
       101
Cotton Kurti
         1        499             25

       102
Floral Saree
         2        799             15

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID     PRICE STOCK_QUANTITY
----------- ---------- --------------

       103
Leggings
         3        299             40

       104
Anarkali Kurti
```

Display selected product details:

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Meesho_Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Cotton Kurti
       499

       102
Floral Saree
       799

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       103
Leggings
```

1.2 WHERE Clause

The WHERE clause is used to filter records based on a specified condition.

Find product ID and name where category ID is 1:

```
SQL> SELECT Product_ID, Product_Name
  2  FROM Meesho_Product
  3  WHERE Category_ID = 1;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
       101
Cotton Kurti

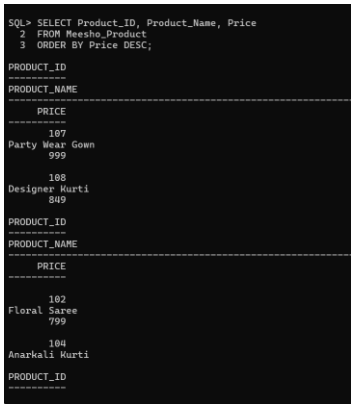
       104
Anarkali Kurti

       108
Designer Kurti
```

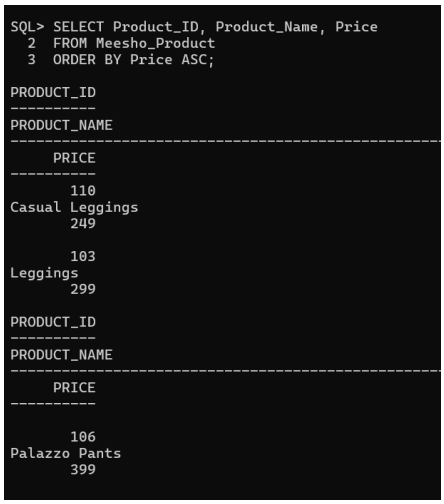
1.3 ORDER BY

The ORDER BY clause is used to sort query results in ascending or descending order.

Display products in descending order of price:



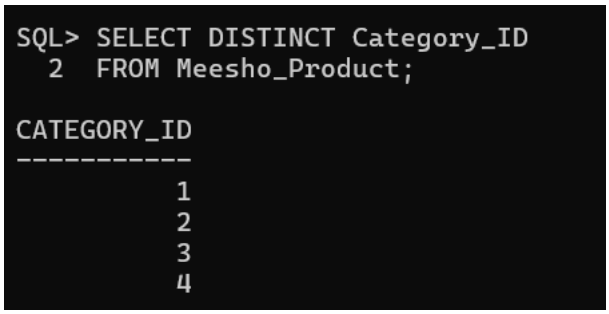
Display products in ascending order of price:



1.4 DISTINCT

The DISTINCT keyword is used to display unique values and eliminate duplicate entries.

Display all unique category IDs:



2. Search Products

2.1 Search Products Based on Price

Products can be searched using their price to identify items within a particular price range.

Display products priced between ₹500 and ₹1500:

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Meesho_Product
  3  WHERE Price BETWEEN 500 AND 1500;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------
       102
Floral Saree
       799

       104
Anarkali Kurti
       699

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------

       105
Printed Saree
       599

       107
Party Wear Gown
```

2.2 Search Products Based on Category

Products can be filtered according to their category.

Display products belonging to category ID 2:

```
SQL> SELECT Product_ID, Product_Name, Category_ID
  2  FROM Meesho_Product
  3  WHERE Category_ID = 2;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID
-----------
       102
Floral Saree
          2

       105
Printed Saree
          2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID
-----------

       109
Cotton Saree
          2
```

3. Retrieve Customer Information

The SELECT statement is used to retrieve customer details from the Customer table.

Display all customers:

```
SQL> SELECT * FROM Customer;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE_NUMBER
---------------
ADDRESS
--------------------------------------------------------------------------------
       101
Priya
priya@gmail.com

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE_NUMBER
---------------
ADDRESS
--------------------------------------------------------------------------------
9876543210
Chennai
```

Display customer ID, name, and email:

```
SQL> SELECT Customer_ID, Customer_Name, Email
  2  FROM Customer;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------
EMAIL
--------------------------------------------------------------
       101
Priya
priya@gmail.com

       102
Divya
divya@gmail.com

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------
EMAIL
--------------------------------------------------------------
```

4. Apply Filtering Conditions

Multiple conditions can be combined using AND, OR, and NOT operators to filter records.

Display products priced below ₹1000 and having stock greater than zero:

```
SQL> SELECT Product_ID, Product_Name, Price, Stock_Quantity
  2  FROM Meesho_Product
  3  WHERE Price < 1000
  4  AND Stock_Quantity > 0;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
       101
Cotton Kurti
       499             25

       102
Floral Saree
       799             15

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------

       103
Leggings
       299             40

       104
Anarkali Kurti
```

Display products belonging to category 1 or category 2:

```
SQL> SELECT Product_ID, Product_Name, Category_ID
  2  FROM Meesho_Product
  3  WHERE Category_ID = 1
  4  OR Category_ID = 2;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID
-----------
       101
Cotton Kurti
          1

       102
Floral Saree
          2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID
-----------

       104
Anarkali Kurti
          1

       105
```

Display products that do not belong to category 1:

```
SQL> SELECT Product_ID, Product_Name, Category_ID
  2  FROM Meesho_Product
  3  WHERE NOT Category_ID = 1;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID
-----------
       102
Floral Saree
           2

       103
Leggings
           3

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID
-----------

       105
Printed Saree
           2

       106
Palazzo Pants
```

5. Basic Business Reports

Business reports help analyse product prices, customer orders, and product sales.

5.1 Product Price Report

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Meesho_Product
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       107
Party Wear Gown
       999

       108
Designer Kurti
       849

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       102
Floral Saree
       799

       104
Anarkali Kurti
```

5.2 Customer Order Report

This report displays customer order details by joining the Customer and Orders tables

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         O.Order_ID,
  4         O.Order_Date,
  5         O.Total_Amount
  6  FROM Customer C
  7  JOIN Orders O
  8  ON C.Customer_ID = O.Customer_ID
  9  ORDER BY O.Order_Date DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- --------- ------------
       101
Priya
       211 23-SEP-26          899

       102
Divya
       202 11-SEP-26         1299

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- --------- ------------
```

5.3 Product Order Report

This report displays products included in customer orders.

```
SQL> SELECT O.Order_ID,
  2         OD.Product_ID,
  3         P.Product_Name,
  4         OD.Quantity,
  5         OD.Price,
  6         OD.Quantity * OD.Price AS Subtotal
  7  FROM Orders O
  8  JOIN Order_Details OD
  9  ON O.Order_ID = OD.Order_ID
 10  JOIN Meesho_Product P
 11  ON OD.Product_ID = P.Product_ID
 12  ORDER BY Subtotal DESC;

  ORDER_ID PRODUCT_ID
---------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE   SUBTOTAL
---------- ---------- ----------
       208        104
Anarkali Kurti
         2        799       1598

       204        105
Printed Saree
         2        749       1498

  ORDER_ID PRODUCT_ID
---------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE   SUBTOTAL
---------- ---------- ----------
```

5.4 Customer-wise Total Orders

This report calculates the total number of orders placed by each customer.

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         COUNT(O.Order_ID) AS Total_Orders
  4  FROM Customer C
  5  LEFT JOIN Orders O
  6  ON C.Customer_ID = O.Customer_ID
  7  GROUP BY C.Customer_ID, C.Customer_Name
  8  ORDER BY Total_Orders DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS
------------
        101
Priya
           2

        102
Divya
           1

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS
------------
```

5.5 Customer-wise Total Purchase Amount

This report calculates the total purchase amount for each customer based on their orders.

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         COALESCE(SUM(O.Total_Amount), 0) AS Total_Purchase_Amount
  4  FROM Customer C
  5  LEFT JOIN Orders O
  6  ON C.Customer_ID = O.Customer_ID
  7  GROUP BY C.Customer_ID, C.Customer_Name
  8  ORDER BY Total_Purchase_Amount DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE_AMOUNT
---------------------
         101
Priya
                 1798

         102
Divya
                 1299

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE_AMOUNT
---------------------


SQL>
```

Conclusion

The SQL queries demonstrate how to retrieve, filter, sort, and analyse data in the Meesho Women's Fashion E-Commerce Database Management System. These queries help manage product details, customer information, and order-related business reports efficiently.